PHYS 230 - Electricity and Magnetism
Lecture 18

Instructor: Saeed Rastgoo

UNIVERSITY OF ALBERTA

Outline

4. Capacitance and Dielectrics
   - Capacitors in Series vs. Parallel Connection
   - Energy Stored in a Capacitor

All figures in these slides are from one of the following sources, unless cited otherwise:

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Series Connection and Equivalent Capacitance

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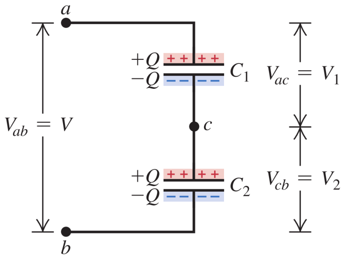

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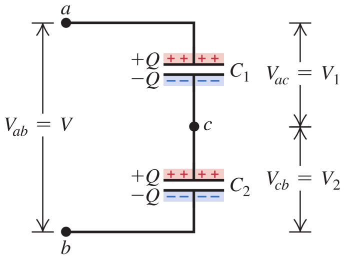

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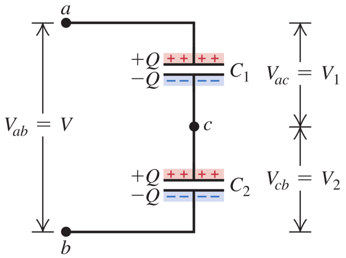

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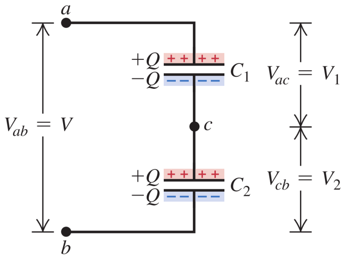

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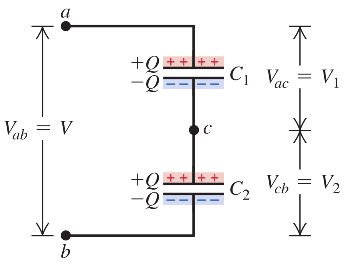

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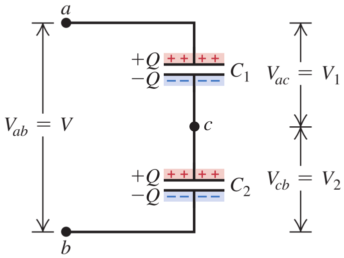

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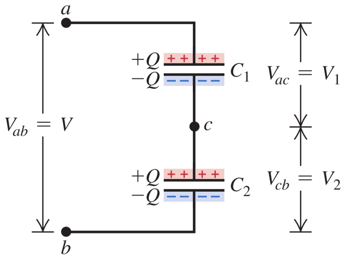

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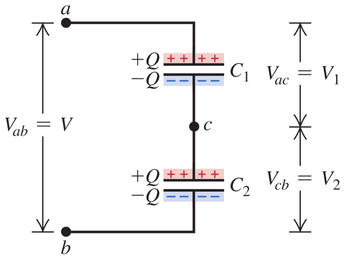

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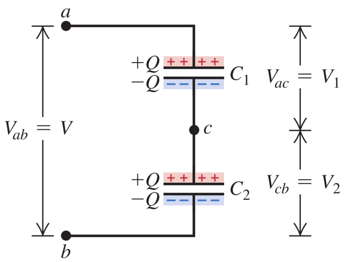

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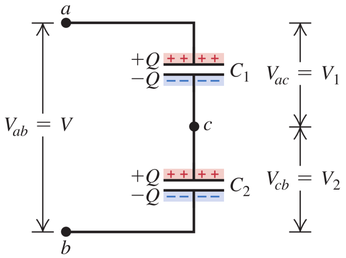

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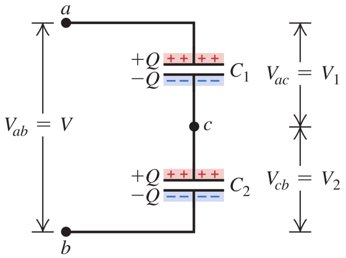

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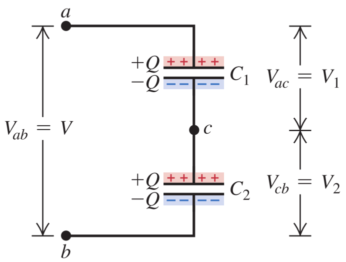

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Series Connection and Equivalent Capacitance

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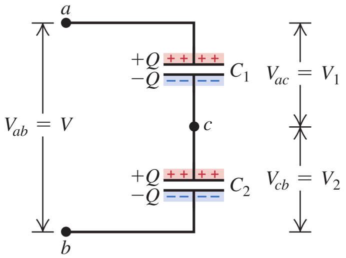

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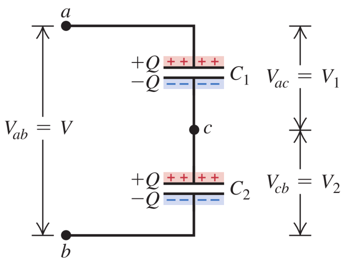

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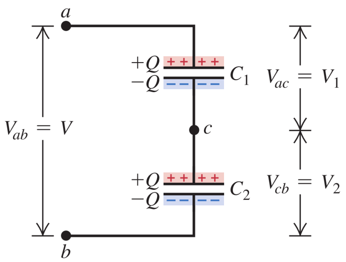

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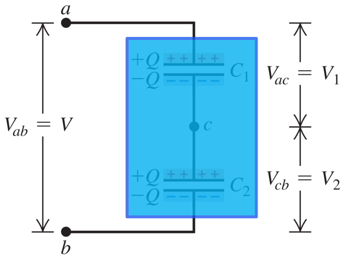

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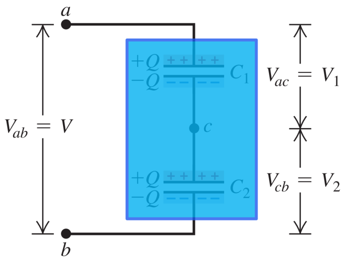

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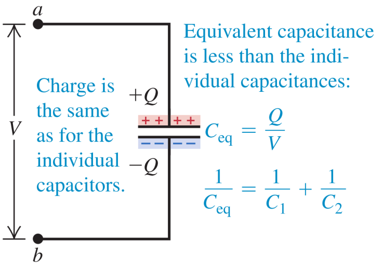

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Two capacitors with capacitance C_1 and C_2 connected in series are equivalent to a single capacitor with

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- potential difference

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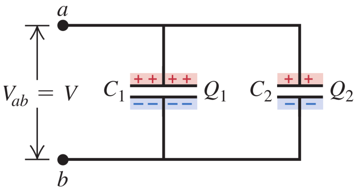
- capacitance

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} \Rightarrow C = \frac{C_1 C_2}{C_1 + C_2}$$

  is always smaller than the individual capacitances

Parallel Connection and Equivalent Capacitance

For capacitors in **parallel** connection we have



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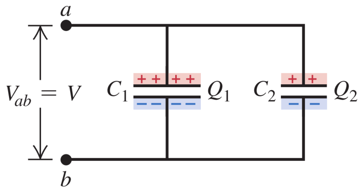

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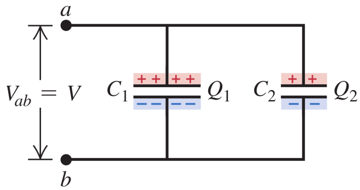

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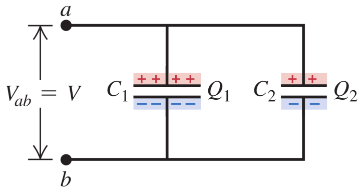

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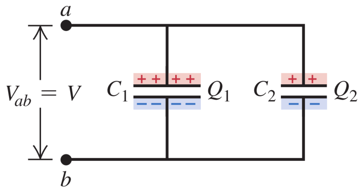

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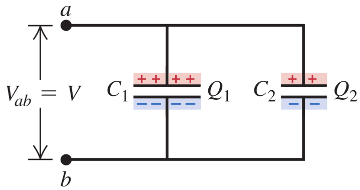

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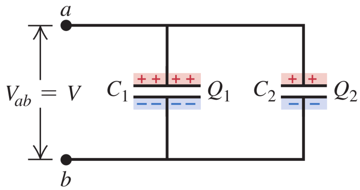

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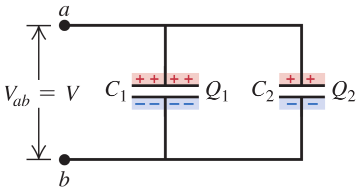

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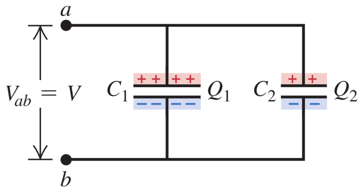

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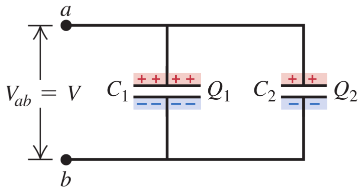

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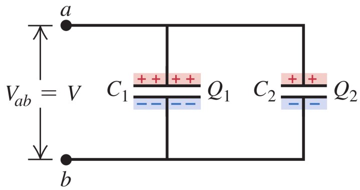

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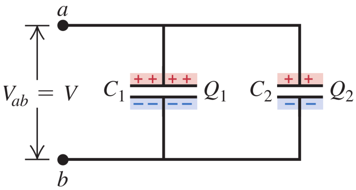

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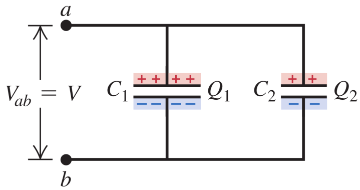

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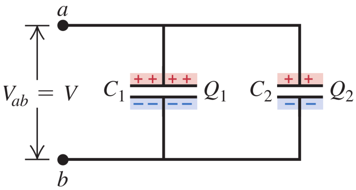

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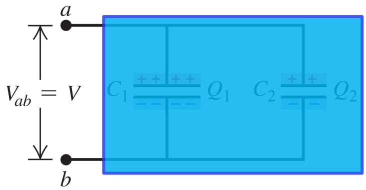

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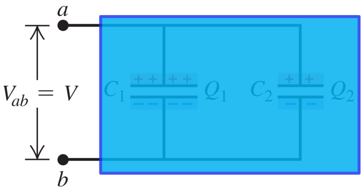

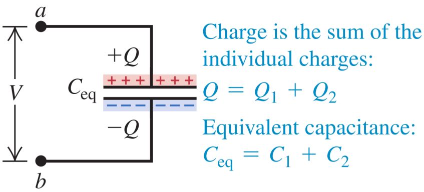

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Two capacitors with capacitance C_1 and C_2 connected in parallel are equivalent to a single capacitor with

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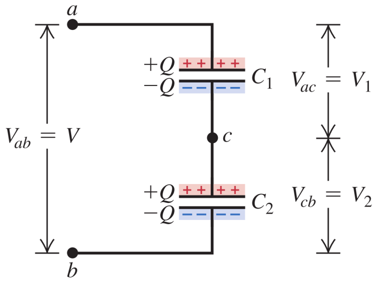
$$C = C_1 + C_2$$
is always larger than the individual capacitances

Exercise 1 https://epoll.srv.ualberta.ca – Code: MBD

In figure $C_1 = 6\,\mu$F and $C_2 = 3\,\mu$F and $V_{ab} = 18$ V. Find the equivalent capacitance, the charge, and potential difference for each capacitor.



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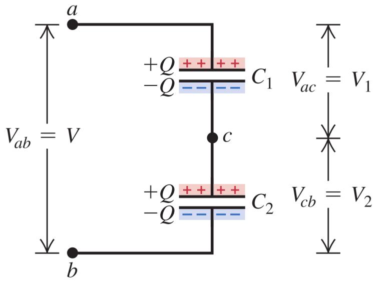

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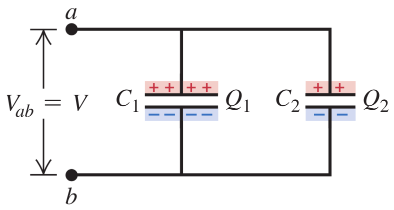

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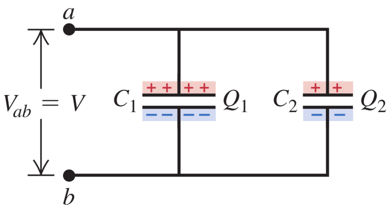
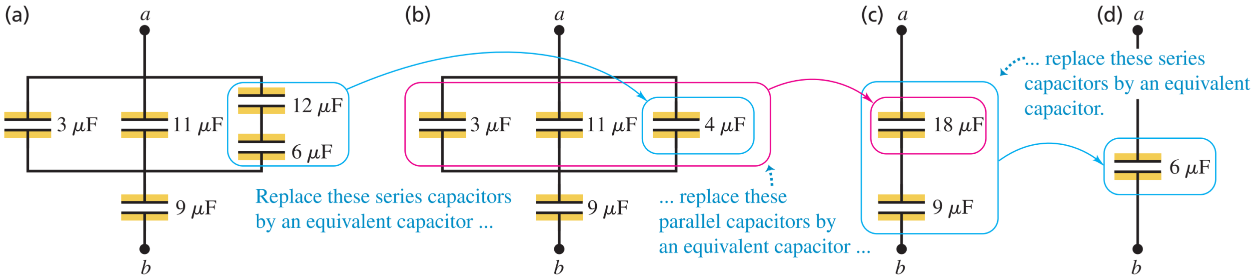


Exercise 3 https://epoll.srv.ualberta.ca – Code: MBD

Find the equivalent capacitance of the five-capacitor network shown in figure.

Figure **24.10** (a) A capacitor network between points *a* and *b*. (b) The 12 μF and 6 μF capacitors in series in (a) are replaced by an equivalent 4 μF capacitor. (c) The 3 μF, 11 μF, and 4 μF capacitors in parallel in (b) are replaced by an equivalent 18 μF capacitor. (d) Finally, the 18 μF and 9 μF capacitors in series in (c) are replaced by an equivalent 6 μF capacitor.



Exercise 3 https://epoll.srv.ualberta.ca – Code: MBD

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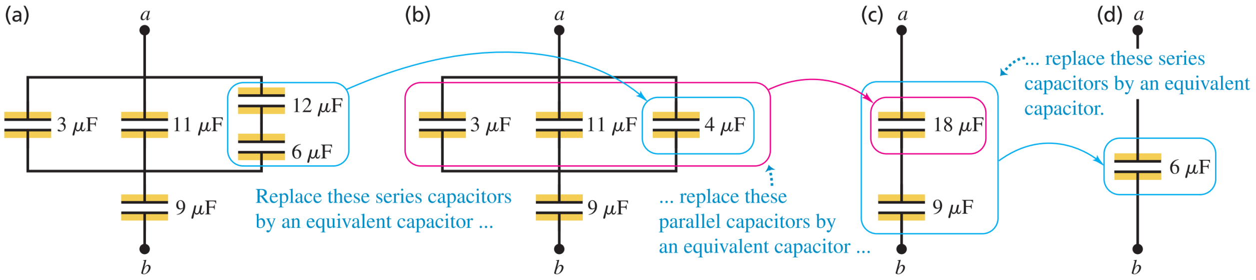

Outline

4. Capacitance and Dielectrics
   - Capacitors in Series vs. Parallel Connection
   - Energy Stored in a Capacitor

All figures in these slides are from one of the following sources, unless cited otherwise:

University Physics With Modern Physics, Young & Freedman
Physics for Scientists and Engineers, Hawkes, et al.
Physics for Scientists and Engineers With Modern Physics, Serway, Jewett

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Although we have derived this relationship for a parallel-plate capacitor only, it turns out to be **valid for any capacitor in vacuum** and indeed **for any electric-field configuration in vacuum**, even if there is no capacitor there.